

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 1

Course Code: CSE276

Course Title: ARTIFICIAL INTELLIGENCE
FOUNDATIONS

Course Outcome: CO1

Paper: Sample Multiple Choice Questions - ANSWER
KEY

Maximum Marks: 40 x 1 = 40

Unit: 1

Unit Title: Foundations of AI and AI Problem Solving

Unit Mapped RBT: L2 (Understand)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (Key)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1	A	Q11	D	Q21	A	Q31	C
Q2	B	Q12	B	Q22	C	Q32	A
Q3	C	Q13	D	Q23	B	Q33	A
Q4	D	Q14	C	Q24	D	Q34	D
Q5	B	Q15	B	Q25	C	Q35	B
Q6	D	Q16	A	Q26	D	Q36	C
Q7	A	Q17	B	Q27	A	Q37	C
Q8	C	Q18	A	Q28	B	Q38	B
Q9	C	Q19	D	Q29	D	Q39	D
Q10	A	Q20	C	Q30	B	Q40	A

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the capability class of Artificial Intelligence that describes a system competent at exactly one task.

- A) Artificial Narrow Intelligence <-- correct
- B) Artificial General Intelligence
- C) Artificial Super Intelligence
- D) Artificial Reactive Intelligence

Answer: A. Every system in service today is narrow: it performs one bounded task and cannot transfer that competence to another.

Q2. Identify the functionality class of a system that uses recent past observations as well as the present one, such as a self-driving car.

- A) Reactive machine
- B) Limited memory <-- correct
- C) Theory of mind
- D) Self-aware

Answer: B. A limited-memory system retains a short history, which is what lets a vehicle judge the speed and direction of the traffic around it.

Q3. State the component of a problem formulation that decides whether a state is a solution.

- A) The initial state
- B) The path cost
- C) The goal test <-- correct

D) The transition model

Answer: C. The goal test is applied to a state and answers yes or no. It may name one particular state or describe a property that any acceptable state must have.

Q4. Name the quantity that reports the average number of successors a state has.

- A) The path cost
- B) The solution depth
- C) The state count
- D) The branching factor <-- correct

Answer: D. The branching factor, written b , together with the depth d is what fixes the size of the tree an uninformed search must explore.

Q5. Identify the performance measure that reports whether a search returns the cheapest of the available solutions.

- A) Completeness
- B) Optimality <-- correct
- C) Time complexity
- D) Space complexity

Answer: B. Completeness asks only whether a solution is found at all. Optimality is the stronger property of finding the best one.

Q6. Name the field that combines statistics, programming and domain knowledge to obtain insight from data.

- A) Deep learning
- B) Expert systems
- C) Robotics
- D) Data science <-- correct

Answer: D. Data science overlaps with machine learning but is broader, taking in data collection, cleaning, analysis and the communication of findings.

Q7. State the phase of the Artificial Intelligence lifecycle in which the performance of a running system is tracked.

- A) Monitoring <-- correct
- B) Problem definition
- C) Data preparation
- D) Model development

Answer: A. Monitoring detects the decay in performance that follows when the incoming data drifts away from what the system was built on.

Q8. Identify the capability class that would match a human being across arbitrary tasks.

- A) Artificial Narrow Intelligence
- B) Reactive machine intelligence
- C) Artificial General Intelligence <-- correct
- D) Limited memory intelligence

Answer: C. General intelligence would carry competence from one domain to another at roughly human breadth. No such system presently exists.

Q9. Name the component of a problem formulation that fixes where the search begins.

- A) The goal test
- B) The action set
- C) The initial state <-- correct
- D) The path cost

Answer: C. The initial state is the configuration the agent starts from, and every state the search reaches is derived from it by applying actions.

Q10. State the performance measure that reports how much memory a search algorithm requires.

- A) Space complexity <-- correct
- B) Time complexity
- C) Completeness
- D) Optimality

Answer: A. Space complexity often decides usability in practice, since a search that would eventually finish is of no use if it exhausts memory first.

Q11. Identify the subfield that uses multi-layer neural networks to learn its own features from raw input.

- A) Expert systems
- B) Data science
- C) Symbolic reasoning
- D) Deep learning <-- correct

Answer: D. Deep learning removes much of the need for hand-designed features, which is what separates it from the classical machine learning before it.

Q12. Name the function that records the total cost of a sequence of actions.

- A) The heuristic function
- B) The path cost <-- correct
- C) The goal test
- D) The branching factor

Answer: B. The path cost accumulates the cost of each step along the path, and it is the quantity an optimal search sets out to minimise.

Q13. State the sector in which Artificial Intelligence is applied to yield prediction and to pest identification from leaf photographs.

- A) Manufacturing
- B) Finance
- C) Education
- D) Agriculture <-- correct

Answer: D. These are the standard agricultural applications named on the syllabus, alongside soil assessment and irrigation planning.

Q14. Name the set of every state that can be reached from the initial state by applying actions.

- A) The solution space
- B) The goal set
- C) The state space <-- correct
- D) The frontier

Answer: C. The state space is everything reachable. The solution space is the far smaller collection of states or paths that satisfy the goal test.

Q15. Identify the phase of the Artificial Intelligence lifecycle in which the raw data is cleaned and made ready for use.

- A) Deployment
- B) Data preparation <-- correct
- C) Monitoring
- D) Problem definition

Answer: B. Preparation covers missing values, inconsistent categories, scaling and encoding, and it usually consumes more of the project than the modelling does.

Q16. State the sector in which Artificial Intelligence is applied to read chest X-rays in support of a radiologist.

- A) Healthcare <-- correct
- B) Smart cities
- C) Agriculture
- D) Education

Answer: A. Diagnostic imaging is among the most developed healthcare applications, used to support rather than replace the clinician's judgement.

Q17. Explain why deep learning is described as a subset of machine learning.

- A) It was invented before machine learning was
- B) It is one family of methods for learning from data <-- correct
- C) It requires no data of any kind to be trained
- D) It applies only to problems written as rules

Answer: B. Machine learning is any approach that learns behaviour from data; deep learning is the particular branch that does so with many-layered neural networks.

Q18. Distinguish a reactive machine from a limited-memory system.

- A) A reactive machine keeps no record of earlier input <-- correct
- B) A limited-memory system keeps no record of input
- C) Both of the two keep a full record of all input
- D) Neither of the two responds to its current input

Answer: A. A reactive machine maps the present situation straight to an action. A limited-memory system also draws on a short history of what it has recently observed.

Q19. Describe what a performance measure contributes to the comparison of two search algorithms.

- A) A guarantee that both will find a solution
- B) A reduction in the size of the state space
- C) A heuristic estimate of the remaining cost
- D) A common basis on which their behaviour can be judged <-- correct

Answer: D. Completeness, optimality, time and space complexity let two methods be compared without running them, and they often disagree, forcing a trade-off.

Q20. Interpret the following as a functionality class: a chess program that evaluates only the position in front of it and keeps no record of the moves already played.

- A) A limited-memory system
- B) A theory-of-mind system
- C) A reactive machine <-- correct
- D) A self-aware system

Answer: C. It responds to the present state alone. Deep Blue is the classic example of a powerful system that is nonetheless purely reactive.

Q21. Predict the effect on an exhaustive search of increasing the depth of the shallowest solution by one.

- A) The number of states examined is multiplied by the branching factor <-- correct
- B) The number of states examined is increased by one
- C) The number of states examined is left unchanged
- D) The number of states examined is halved

Answer: A. The tree grows as b raised to the depth, so each additional level multiplies the work by b . This is why uninformed search becomes impractical so quickly.

Q22. Recognize the application area described: sensors and cameras adjust the timing of traffic signals across a city so that waiting time falls.

- A) Healthcare
- B) Manufacturing
- C) Smart cities <-- correct
- D) Education

Answer: C. Adaptive traffic control, along with waste routing and energy management, is among the standard smart-city applications on the syllabus.

Q23. Explain why a problem has to be well defined before any search technique can be applied to it.

- A) The algorithm needs a labelled training dataset
- B) The algorithm needs states, actions, a goal test and costs <-- correct
- C) The algorithm needs the answer to be known first
- D) The algorithm needs the problem to be very small

Answer: B. A search method manipulates a formal structure. Without the components of a formulation there is nothing for it to expand, test or compare.

Q24. Summarize the difference between Artificial Intelligence and data science.

- A) Data science builds systems that act, AI produces insight
- B) The two terms describe exactly the same activity
- C) AI uses data while data science never uses data
- D) AI builds systems that act, data science produces insight <-- correct

Answer: D. The two overlap heavily in their methods, but the aim differs: an AI system takes decisions or actions, while data science informs the decisions people take.

Q25. Distinguish a decomposable problem from one that is not.

- A) A decomposable problem cannot be split at all
- B) A decomposable problem has exactly one solution
- C) A decomposable problem splits into independent subproblems <-- correct
- D) A decomposable problem needs no goal test

Answer: C. When the parts can be solved separately and combined, the work falls dramatically. Where the parts interact, the problem must be tackled as a whole.

Q26. Explain the difference between a problem whose answer is a state and one whose answer is a path.

- A) One needs a heuristic, the other needs none at all
- B) One has a goal test, the other has no goal test
- C) One is deterministic and the other is stochastic

D) One needs only the final configuration, the other the steps <-- correct

Answer: D. In the eight-queens problem only the final arrangement matters; in route planning the sequence of moves is the answer, and its cost is what is minimised.

Q27. Interpret a problem in which the same action taken in the same state can lead to different outcomes.

A) Its outcome is not predictable with certainty <-- correct

B) Its state space must be infinitely large

C) Its goal test cannot be written down

D) Its solution must always be a single state

Answer: A. Unpredictable outcomes force the agent to plan for several possibilities or to replan as it goes, rather than committing to one sequence in advance.

Q28. Describe what the solution space of a problem contains.

A) Every state reachable from the initial state

B) Every state or path that satisfies the goal test <-- correct

C) Every action available at the initial state

D) Every heuristic that could be applied to it

Answer: B. The solution space is a subset of the state space, and it is usually a very small one, which is why a search is needed to locate its members.

Q29. Explain why Artificial General Intelligence has not yet been achieved.

A) No computer is fast enough to run any AI method

B) No organisation has attempted to build such a system

C) No mathematical definition of intelligence exists

D) No system transfers competence across arbitrary domains <-- correct

Answer: D. Present systems are trained for bounded tasks and degrade sharply outside them. Breadth of transfer, not raw performance, is the missing property.

Q30. Recognize the application area described: an assistant builds a personalised revision plan from a student's assessment record.

A) Finance

B) Education <-- correct

C) Robotics

D) Agriculture

Answer: B. Adaptive learning, automated feedback and personalised revision are the education applications named on the syllabus.

Q31. Distinguish a machine learning system from a rule-based one by the origin of the rules each uses.

A) A rule-based system infers its rules from data

B) Both of the two infer their rules from data

C) Learning infers the rules, a rule-based system is given them <-- correct

D) Both of the two are given their rules by hand

Answer: C. A rule-based system encodes what a specialist can articulate. Learning recovers the pattern from examples, which is what allows tasks nobody can state as rules.

Q32. Explain why the way a problem is represented can matter more than the algorithm chosen to solve it.

A) A good representation removes most of the search <-- correct

B) A representation fixes the branching factor at one

C) Every algorithm behaves identically on any problem

D) The representation decides the memory available

Answer: A. Choosing states and actions that exclude pointless configurations can shrink a space by orders of magnitude, which no algorithm can compensate for afterwards.

Q33. Predict the difficulty of applying Artificial Intelligence in a sector that records very little data.

A) Learned methods struggle, so rules or expert input are needed <-- correct

B) Learned methods perform better on small datasets

C) The sector cannot use any form of automation

D) The lifecycle no longer requires a monitoring phase

Answer: A. Data-driven methods need examples. Where these are scarce, knowledge-based approaches, transfer from a related domain, or plain rules remain the practical route.

Q34. Summarize what the characteristics of a problem tell a designer before a method is chosen.

A) How many people will use the finished system

B) Which programming language should be used

C) How much the finished system will cost to run

D) Whether it can be decomposed, undone and predicted <-- correct

Answer: D. Decomposability, recoverability, predictability of outcome and the form of the answer between them rule most techniques in or out before any code is written.

Q35. Describe what makes a task better suited to Artificial Intelligence than to ordinary programming.

A) The task has to be completed very quickly

B) The rules cannot be written down in advance <-- correct

C) The task involves a large volume of storage

D) The task must run on specialised hardware

Answer: B. Where the variation is too great to enumerate, or the experts cannot state the rules they follow, the behaviour has to be learned or reasoned rather than coded.

Q36. Explain how Artificial Intelligence is applied in manufacturing.

- A) In writing the payroll of the factory staff
- B) In storing the production records securely
- C) In visual inspection and in predicting equipment failure <-- correct
- D) In printing the labels for finished goods

Answer: C. Defect detection from images and predictive maintenance from sensor histories are the two standard manufacturing applications, alongside robotic assembly.

Q37. Explain why the success criteria of an Artificial Intelligence project should be agreed before development begins.

- A) They are required by the programming language
- B) They fix the branching factor of the search
- C) They decide what is built and how it will be judged <-- correct
- D) They determine which sector the system serves

Answer: C. Without an agreed measure the project cannot be shown to have succeeded, and disputes about what the system was for surface only after it is built.

Q38. Interpret the claim that a system belongs to the self-aware class of the functionality classification.

- A) It is the class that every modern system occupies
- B) It is a theoretical class that no system occupies <-- correct

- C) It means the system keeps a log of its own actions
- D) It means the system can be switched off remotely

Answer: B. Theory of mind and self-awareness are hypothetical stages in the classification. Everything deployed today is reactive or limited memory.

Q39. Distinguish an irrecoverable problem from a recoverable one.

- A) A recoverable problem allows no action to be undone
- B) Both of the two allow every action to be undone
- C) Neither of the two allows any action to be undone
- D) An irrecoverable problem allows no action to be undone <-- correct

Answer: D. In an irrecoverable problem a wrong move cannot be repaired by backtracking, so the method must look ahead carefully before committing to anything.

Q40. Predict what happens to a search on an infinite state space when no depth limit is imposed.

- A) It may follow one branch forever and never terminate <-- correct
- B) It reports that no solution exists at all
- C) It is guaranteed to find the shallowest solution
- D) It automatically switches to a heuristic method

Answer: A. Completeness fails on an infinite space unless the search is bounded or explores level by level, which is the difficulty depth-limited search addresses.

----- End of Sample Question Set -----